National University of Computer and Emerging Sciences, Lahore Campus

Course:	**Software Testing**	**Course Code:**	**CS497**
Program:	**BS (CS)**	**Semester:**	**Spring 2020**
Duration:	**60 Minutes (1 Hour)**	**Total Marks:**	**30**
Paper Date:	**26-Feb-20**	**Weight**	**10%**
Section:	**C & D**	**Page(s):**	**5**
Exam:	**Mid I**	**Roll No:**	

Instruction/Notes:

1. Attempt all questions on the question paper in context of class discussions, handouts, text book(s) etc.
2. State your assumptions clearly.
3. <u>Do not submit any extra sheet, it will not be graded.</u>

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Question 1 **5+5+5 = 15 Marks**

The following program (with line numbers mentioned) computes x to the power of y, where x and y are integers.

INPUT: The x and y values.

OUTPUT: x raised to the power of y is printed.

<u>Code:</u>

```
1  void pow ( int x, int  y)
2 {
3     float z;
4int p;
5     if (y < 0)
6         p = 0 – y;
7else
8p = y;
9z = 1.0;
10while (p != 0)
11    {
12        z = z * x;
13  p = p – 1;
14    }
15if (y < 0)
16z = 1.0 / z;
17    printf(z);
18 }
```

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a. Draw dataflow graph (definition and use nodes for variables clearly annotated).

b. Identify paths to satisfy **all DU path** criteria w.r.t. to variable 'x' and 'y'.

c. Identify/Highlight DCF(definition clear path in each DU path.)

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Question 2 **5+5+5 = 15 Marks**

Requirement:

The system reads in three positive values from the standard input. The three values A,B, and C are interpreted as representing the lengths of the sides of a triangle. The system then prints a message on the standard output saying whether the triangle is scalene, isosceles, equilateral, or right angled if a triangle can be formed, if the triangle cannot be formed then it would display "Not a triangle".

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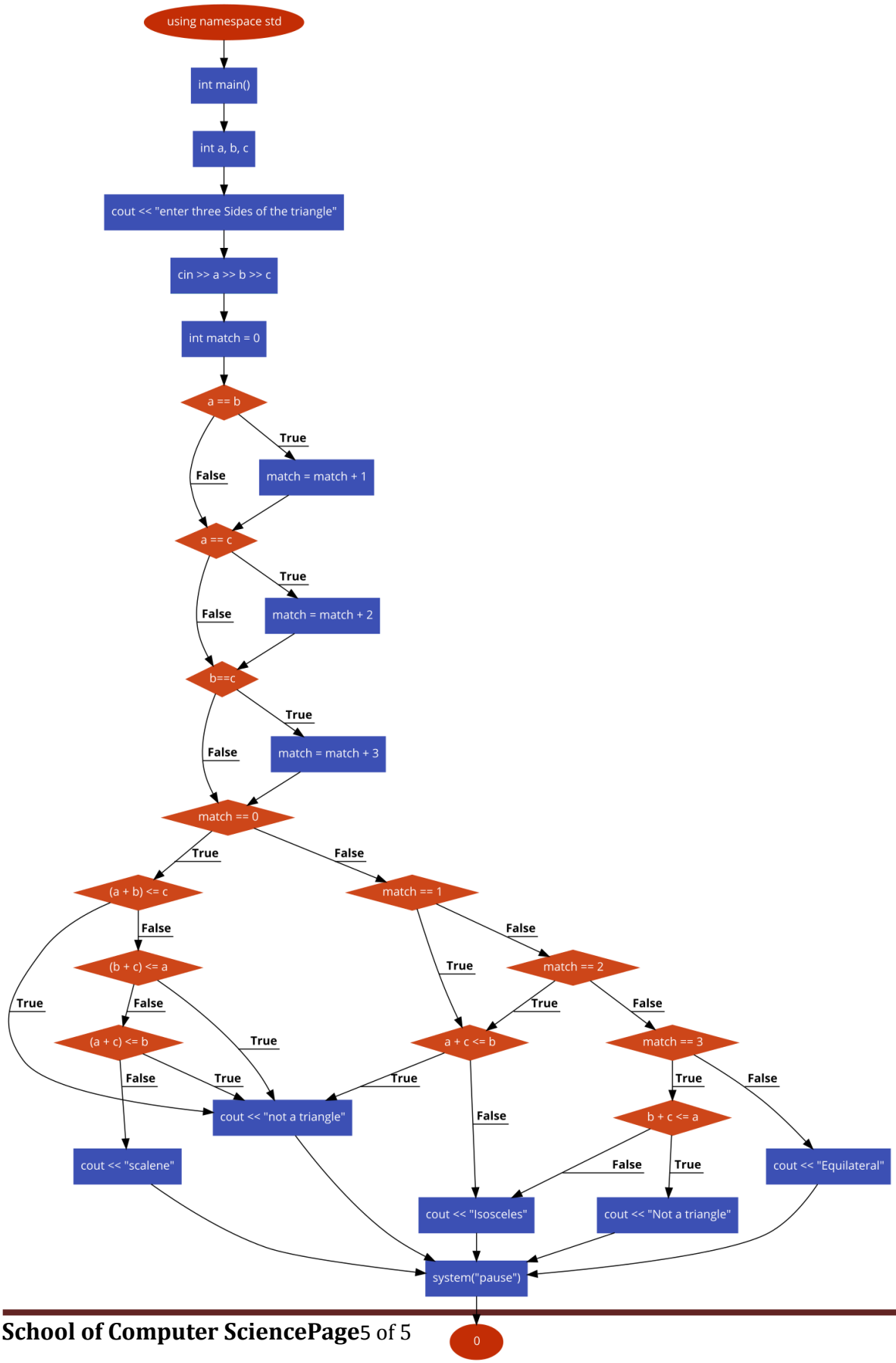

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A program is developed and above control flow graph is generated for that code to satisfy the requirement. Answer the following questions for the above control flow graph.

a. What is the minimum number of test cases required to perform basis path testing of the program? Show the complete process of finding the number of required test cases. Use more than one methods to validate your answer.

b. Identify and list the basis paths in the following table

Sr.	Basis Path

c. Develop test cases for any five basis paths from part (b).

Test Case Id	_____	_____	_____	_____